

Deliverable 1: Refined Scope and Entity Identification

Introduction

The goal is to create a healthcare knowledge graph with a primary focus on Lung Cancer, integrating its symptoms, treatments, diagnostic tests, and medications. This graph will provide insights for clinical decision-making, patient education, and research.

Scope

The scope of the knowledge graph includes:

Diseases:

Central focus: Lung Cancer with subtypes like Non-Small Cell Lung Cancer, Metastatic Lung Cancer, Pleural Effusion, and Pulmonary Fibrosis.

Symptoms:

Cough, Shortness of Breath, Chest Pain, Fatigue, etc.

Treatments:

Chemotherapy, Immunotherapy, Radiation Therapy, Surgery.

Medications:

Drugs like Cisplatin, Atezolizumab, Nivolumab, Docetaxel.

Diagnostic Tests:

Procedures like Biopsy, CT Scan, MRI.

Lifestyle Changes:

Quit Smoking, Healthy Diet, Regular Exercise.

Entities and Relationships

This knowledge graph establishes the following entities and relationships:

Entities:

Diseases, Symptoms, Treatments, Medications, Diagnostic Tests, Lifestyle Changes.

Relationships:

1. Disease-HasSymptom: E.g., Lung Cancer → HasSymptom → Cough.
2. Disease-CanBeTreatedWith: E.g., Lung Cancer → CanBeTreatedWith → Chemotherapy.
3. DiagnosticTest-Indicates: E.g., Biopsy → Indicates → Lung Cancer.
4. Treatment-UsesMedication: E.g., Chemotherapy → UsesMedication → Cisplatin.

The schema includes:

Classes: Disease, Symptom, Treatment, Medication, Diagnostic Test, Lifestyle Change.

Attributes:

Disease: Name, Severity, Common Symptoms.

Symptom: Description, Severity, Duration.

Treatment: Name, Effectiveness, Side Effects.

Medication: Name, Dosage, Drug Interactions.

Relationships:

1. Disease → HasSymptom → Symptom
2. Disease → CanBeTreatedWith → Treatment
3. Treatment → UsesMedication → Medication

This schema was adopted and refined based on Schema.org's MedicalEntity vocabulary, ensuring compatibility with ontology standards.

Conclusion

This deliverable defines the scope and foundational structure for the healthcare knowledge graph. The focus on Lung Cancer and its related entities ensures that the graph will provide meaningful insights and support for healthcare applications.